STUDENT CLASSROOM GUIDE

Sampling: does more mean better?

Can a larger sample still give a misleading picture of the whole school?

45 minutes • Junior secondary • pairs or groups of 2–4

Before you begin

Frequency, percentages and reading a bar chart. No programming required.

> An authored 45-minute sequence using an existing synthetic inquiry. The 200-person school and its activity group are invented, not a real school survey.

Synthetic starting counts: the whole school has 200 people, with 100 choosing A; the activity group has 40 people, with 36 choosing A. These fixed counts support a paper comparison and are not a newly drawn sample.

One-class sequence

5 min **01. Predict before asking**

Will asking all 40 group members be closer to the school’ s 50% than asking 20 randomly chosen pupils? Explain before running.

12 min **02. Keep size, change who is asked**

Use the selected seed (default 2026) to compare school-20 and club-20. Record the A count, sample size and percentage. Save both pieces of evidence in the original tool.

10 min **03. Increase size within each frame**

Compare club-40 and school-40 with the earlier cases. Which pupils are still excluded by asking only the group?

10 min **04. Draw again and revise**

Use the next-seed control and inspect the listed 30 repeats. Revise “more people always means more accurate”. Do not pick only the nicest result.

8 min **05. Plan a lunch survey**

A pupil asks only the basketball team about favourite lunches. Propose a better sampling frame, note possible non-response, then answer the exit question.

Settings to apply manually

This form updates this classroom page and its share link only. It never loads settings into the activity or changes a saved draft.

Sample size

20

Sampling frame

Whole school, random

Chart axis starts at (%)

0

Experiment seed

2026

Keep this evidence

- Two saved comparisons with method, size, seed and A percentage.
- A revised claim and a lunch-survey plan with one limitation.

> Use Save evidence, Download record and Print in the original inquiry. Its local record stays in this browser; this classroom page does not read, save or import it. Paper responses can be kept separately.

Exit question

If the complete activity group gives 90%, can we claim the whole school is 90%? Why?

Printable worksheet

Write on a printed copy or keep notes separately. This page has no answer storage. Use the original activity to save digital work.

My prediction and reason:

Case / who was asked / size / seed / A count / A percentage:

What changed? What stayed fixed?

My revised claim, evidence and limitation:

Lunch-survey plan and exit response:

Activity source

Sequence adapted from the existing Year 7 sampling inquiry (content g7-1), including its four comparison cases, layered hints and transfer question. The 45-minute pacing is an editorial lesson plan, not a measured learning outcome.

This guide：https://skckenneth.github.io/teaching/classroom-packs/sampling-bias/
Original activity：https://skckenneth.github.io/math-studio/